

Beyond Accuracy: A Cost-Sensitive Framework for Optimal Abstention in Defense-Analytic Classifiers

Precision-Recall-Based Thresholding: Balancing Recall and Precision for Operational Tuning

Precision-Recall-based thresholding represents a widely adopted approach for tuning classification models, particularly in scenarios characterized by class imbalance and asymmetric misclassification costs ²⁵³. At its core, this methodology involves adjusting the probability cutoff used to assign a class label, moving beyond the default 0.5 threshold common in many libraries ³¹⁴. The primary tool for this process is the Precision-Recall Curve (PRC), which graphically depicts the trade-off between precision—the proportion of true positive predictions among all instances predicted as positive—and recall—the proportion of true positive predictions among all actual positive instances across a range of probability thresholds ^{5 11}. For a multi-class problem like the satellite hotspot classifier, this typically involves creating one-vs-rest (or one-vs-all) binary classifiers, where each class is treated as the positive class while all others are grouped into the negative class, resulting in a separate curve for each of the four classes ³⁰⁹³²⁹. The area under this curve (AUPRC) serves as a single scalar metric summarizing performance, with a higher value indicating a better balance between precision and recall ³⁴¹.

The principal strength of the PR-based approach lies in its direct alignment with the operational requirement to manage false-negative risk. In the context of satellite anomaly detection, missing a genuine wildfire is a critical failure with potentially severe consequences, making recall for the 'wildfire' class the paramount concern ^{91 119}. By analyzing the PRC for the 'wildfire' class, an operator can visually inspect the curve to select a threshold that prioritizes high recall, even at the expense of lower precision, thereby minimizing the chance of such a catastrophic error ¹³⁴. This intuitive, visual method for navigating the precision-recall trade-off makes it an accessible diagnostic tool for explaining model behavior to evaluators, a crucial aspect for a government demonstration ¹⁹⁴. Furthermore, the use of PR-curve analysis is deeply entrenched in analogous high-stakes fields such as financial fraud detection and healthcare diagnostics,

where imbalanced datasets and the need to identify rare but critical events are common challenges ^{7 45 168}. Its established presence in these domains lends credibility and familiarity to the evaluation process. For instance, in credit card fraud detection, precision is a key indicator of a model's ability to accurately identify fraudulent transactions without overwhelming investigators with false alarms ⁷, while in clinical settings, the PRC helps determine if a model is appropriate for use and aids in selecting an optimal operating point ¹³⁴.

However, despite its strengths, a naive application of PR-based thresholding presents significant limitations when confronted with the full complexity of the specified operational constraints. The most notable weakness is its lack of an explicit mechanism for modeling operational costs. While the PRC effectively illustrates the geometric relationship between precision and recall, it does not inherently account for the different financial, strategic, or safety-related costs associated with various types of errors ¹⁰². For example, a threshold optimized for maximum recall on 'wildfire' might achieve this goal but could simultaneously produce a high number of false positives for the 'industrial thermal source' class. Such an outcome would still be operationally untenable, yet the standard PR-based framework offers no principled way to penalize this type of error or balance it against the recall gain. This focus on a single trade-off per class ignores the holistic cost-benefit landscape required for a rigorous, defensible decision-support system. The approach treats each class in isolation, which can lead to suboptimal global performance by failing to capture inter-class correlations and the overall distribution of uncertainty ⁸.

Another critical limitation is the challenge of managing multiple, conflicting constraints simultaneously. The user has identified three key operational goals: maximize recall for the critical 'wildfire' class, maintain reviewer workload within a strict budget (e.g., an abstention rate below 40%), and presumably control the false positive rate for other classes. A PR-curve-based approach is ill-suited to solve this complex multi-objective optimization problem elegantly. Adjusting the threshold to meet one constraint often disrupts the others. Manually searching for a threshold that satisfies all conditions is a reactive and inefficient process, lacking the mathematical rigor needed for a high-stakes public-sector application. The system lacks a formal objective function that can balance the competing priorities of minimizing risk (false negatives), controlling cost (reviewer workload), and maintaining efficiency (avoiding excessive false positives). Consequently, while PR-based methods provide an excellent starting point for understanding a model's baseline performance characteristics, they fall short as a standalone, comprehensive solution for this specific problem. They serve best as a foundational diagnostic tool rather than a complete decision-making framework.

Cost-Sensitive Reject-Option Classification: A Decision-Theoretic Framework for High-Stakes Abstention

Cost-Sensitive Reject-Option Classification (CSROC) offers a more sophisticated and theoretically grounded framework for determining when a model should abstain from making a prediction. This paradigm extends traditional classification by introducing a third possible action: rejection or abstention ^{56 102}. The core principle of CSROC is to minimize the expected total cost of the decision-making process, which now includes the costs of correct classifications, incorrect classifications (errors), and the act of rejecting the decision for human review ¹⁰². This transforms the problem from a simple accuracy maximization task into a constrained optimization problem that explicitly incorporates operational realities. The decision rule is derived from a cost function defined by three constant costs: $C_{correct}$, C_{error} , and C_{reject} ¹⁰². Typically, this function is normalized so that $C_{correct}=0$ and $C_{error}+C_{reject}=1$, ensuring that the cost of rejection is always less than the average cost of a random prediction, thus making abstention a rational option when uncertainty is high ¹⁰². This approach is particularly powerful because it allows for the assignment of differential costs to different types of misclassifications, reflecting the varying severity of their consequences.

The defining advantage of CSROC is its direct alignment with the user's operational imperatives. It provides a formal mechanism to balance the high risk of a false negative (e.g., misclassifying a 'wildfire' as something else) against the practical constraint of reviewer workload. By assigning a very high cost to a 'wildfire' false negative, the model's objective function is skewed to prioritize recall for this class above all others. Conversely, a moderate cost can be assigned to a 'mining' misclassification, and a low cost to a correct identification. The cost of rejection (C_{reject}) can be calibrated to represent the opportunity cost of human labor, providing a direct link to the reviewer workload constraint ²⁶⁴. The optimal policy is then the one that finds the decision boundary (the rejection threshold) which minimizes the weighted sum of prediction errors and rejection costs, effectively optimizing the entire human-AI collaboration workflow ³³. This integrated framework is well-suited for the complex, multi-objective nature of the satellite detection problem, where decisions cannot be made in a vacuum but must consider downstream impacts on resources and safety.

This decision-theoretic approach is not merely a theoretical construct; it has deep roots in the same high-stakes domains the user wishes to emulate. The concept of a reject option is central to applications in medical triage, where AI systems assist clinicians by flagging cases that require immediate human attention, thereby improving patient safety while

managing limited clinical resources ¹⁸³²²⁹. Studies have shown that AI-assisted triage can reduce radiologist workload by filtering out non-critical cases while ensuring that serious conditions are not missed ¹⁸⁵. Similarly, in financial fraud detection, where misclassifying a fraudulent transaction can lead to significant financial loss and reputational damage, cost-sensitive learning is a standard practice ^{92 96}. Research in this area demonstrates how custom loss functions and cost matrices can be designed to prioritize the detection of rare but high-impact events ^{78 150}. The TRIAGE benchmark, designed to evaluate the ethical decision-making of AI in high-stakes medical dilemmas, further underscores the relevance and maturity of this approach for defense and security contexts where automated decisions carry significant weight ²³². By adopting CSROC, the proposed system aligns itself with established best practices from fields that share the same fundamental tension between predictive power and operational safety.

Despite its advantages, implementing CSROC introduces its own set of challenges, primarily centered on the specification of the cost matrix. Determining the precise numerical values for the costs of different errors and rejections is a non-trivial task that requires deep domain expertise ⁹⁴. While the relative cost of a 'wildfire' being missed is intuitively very high, quantifying the exact monetary or strategic impact is difficult. Likewise, assigning a meaningful cost to a false positive ('industrial' source incorrectly flagged) or calculating the precise opportunity cost of a single human review requires careful consideration and potentially pilot studies or simulations. This contrasts with the more straightforward visual inspection offered by PR curves. Another consideration is the implementation architecture. The reject option can be implemented in several ways: as a separated rejector (a filter acting before the predictor), a dependent rejector (using the predictor's output, like confidence scores, to decide), or an integrated rejector (where rejection is treated as an additional class during training) ¹⁰². Integrated approaches can be more optimal but may require more complex custom loss functions ¹⁸⁴. Finally, there is a risk of over-abstention. If the cost of rejection is set too low relative to the cost of errors, the model may become overly conservative and reject a large portion of inputs, potentially exceeding the reviewer workload capacity and defeating the purpose of automation. Managing adherence to a predefined "abstention budget" is a recognized research challenge in this area ^{54 187}. Therefore, while CSROC provides a superior framework for addressing the user's needs, its successful deployment hinges on careful cost specification and thoughtful implementation.

Comparative Evaluation of PR-based and CSROC Methods

A direct comparison between Precision-Recall-based thresholding and Cost-Sensitive Reject-Option Classification reveals a clear divergence in their philosophies and capabilities, making one demonstrably more suitable for the specified high-stakes, multi-constraint operational environment. The choice between them is not merely a technical preference but a strategic decision about how to model and optimize for operational reality. The table below synthesizes the key differences based on the user's requirements, drawing upon principles from machine learning theory and analogous high-stakes applications.

Feature	Precision-Recall-Based Thresholding	Cost-Sensitive Reject-Option Classification
Primary Optimization Objective	Maximize F1-score or find an optimal balance between precision and recall for a given class ³³⁴ .	Minimize the expected total cost, which is a weighted sum of prediction errors and rejection actions ¹⁰² .
Handling of Multiple Constraints	Poorly suited for simultaneous optimization. Adjusting the threshold to satisfy one constraint (e.g., high recall) often degrades performance on others (e.g., precision, workload) ²⁹⁵ .	Well-suited for multi-objective optimization. The cost matrix allows for explicit balancing of competing priorities, such as maximizing recall for a critical class while respecting an "abstention budget" ²⁶⁴ .
Management of False-Negative Risk	Directly addresses recall, allowing the selection of a threshold that minimizes false negatives for a specific class (e.g., 'wildfire') ¹³⁴ .	Directly addresses risk by assigning a high cost to specific misclassification types, forcing the model to prioritize avoiding those costly errors during optimization ⁹¹ .
Management of Reviewer Workload	Managed reactively through manual threshold adjustment after model training. No principled mechanism exists to guarantee adherence to a specific abstention rate ¹⁵⁷ .	Managed proactively via the cost function. The cost of rejection (C_{reject}) acts as a parameter that directly controls the model's propensity to abstain, enabling the definition of an "abstention budget" ¹⁸⁷ .
Explainability and Auditability	Good. The Precision-Recall Curve (PRC) provides an intuitive visualization of performance trade-offs, enhancing transparency ¹⁹⁴ . However, the reason for a specific abstention event is not explicitly modeled.	Better. The decision to abstain is explicitly justified by the underlying cost-benefit calculation. Full auditability requires documenting and defending the chosen cost matrix, but the logic is mathematically sound ¹³⁰ .
Alignment with Analogous Fields	Widely used and understood in fraud detection and medical diagnosis, lending it familiarity ⁷ ¹⁶⁸ .	Core concept in medical triage and financial fraud, where the decision to defer to a human expert is a fundamental part of the workflow ¹⁶⁵ ¹⁸³ .

The most salient distinction lies in their approach to optimization. PR-based methods operate on a post-hoc tuning basis, seeking a single probability threshold that optimizes a secondary metric like the F1-score, which is a harmonic mean of precision and recall ³³⁴. This is a powerful technique for understanding a model's inherent trade-offs but is fundamentally a descriptive tool. It describes what the model *can* do at different thresholds. In contrast, CSROC is a prescriptive framework. It integrates the operational goal directly into the learning or decision-making process. It asks not just "what is the

best threshold?" but "what is the optimal policy that minimizes our total operational cost, given our resource constraints?". This shift from descriptive analysis to prescriptive optimization is critical for a system intended for a high-stakes national security application. The latter requires a level of rigor and justification that goes beyond simply reporting a model's performance on a curve.

Furthermore, CSROC offers a more robust mechanism for handling holistic uncertainty. The one-vs-rest strategy employed in multi-class PR-curve analysis evaluates each class independently, which can be misleading ⁸. An instance might be confidently classified as 'not-wildfire' while still being ambiguous enough to belong to any of the other three classes. A PR-based approach focused solely on the 'wildfire' class might fail to reject this borderline case, whereas an integrated CSROC model, which considers the full probability distribution across all classes, is better equipped to recognize such ambiguity and trigger a rejection ¹⁰². This is particularly relevant in anomaly detection, where the goal is often to identify patterns that deviate significantly from normal behavior, a state that may not map cleanly to any single predefined class. The concept of novelty rejection, where the model identifies inputs that are sufficiently different from the training data, is naturally incorporated into more advanced rejection frameworks and is closely related to uncertainty quantification ^{53 102}.

Finally, the issue of auditability is addressed more directly by CSROC. While the visual appeal of a PRC is useful for a dashboard narrative, it ultimately obscures the reasoning behind individual abstention decisions. A defense evaluator will require a more defensible rationale. With CSROC, the reason for an abstention is explicit: the model determined that the expected cost of predicting a class was higher than the cost of deferring to a human expert. To make this auditable, the team would need to clearly document and justify the cost matrix used, perhaps linking the high cost assigned to a 'wildfire' false negative to established risk assessments or historical incident data. This creates a transparent and traceable chain of reasoning from operational goals to final model decisions, a cornerstone of trustworthy AI in the public sector ¹²⁹²⁰⁸. The PR-based approach, while simpler to implement, struggles to provide this depth of justifiable logic.

Practical Implementation Workflow and Operationalization for a Public-Sector Demo

To successfully address the research goal within the constraints of a tight timeline and a high-stakes demonstration, a pragmatic, hybrid workflow is recommended. This approach leverages the strengths of both PR-based analysis and CSROC, using the former as a diagnostic tool and the latter as the foundation for the final decision-making system. This workflow ensures technical robustness, operational soundness, and the necessary auditability for a national security audience. The process begins with a critical prerequisite step and proceeds through analysis, cost modeling, implementation, and validation.

First, **Step 0: Probability Calibration** is a mandatory preliminary step. Before any thresholding or rejection logic is applied, the raw output probabilities from the CatBoost model must be calibrated. Uncalibrated probabilities are often poorly aligned with true likelihoods, meaning a model outputting a probability of 0.8 for a class is not necessarily correct 80% of the time ¹⁷⁸. Methods like Platt scaling or temperature scaling can be used to transform the model's outputs into well-calibrated probabilities, which are essential for both PR-based analysis and the cost calculations in CSROC ¹⁷³. Without this step, the confidence scores used for thresholding would be unreliable, undermining the entire system. CatBoost itself supports various evaluation metrics, and tools exist to assess and improve calibration ²⁰³²⁰⁵.

Second, **Step 1: Diagnostic Analysis with Multi-Class Precision-Recall Curves**.

Following calibration, the next step is to generate multi-class Precision-Recall curves for the trained CatBoost model ³²⁹. This analysis serves two purposes. First, it provides a baseline understanding of the model's inherent performance trade-offs for each of the four classes ('wildfire', 'industrial', 'mining', 'agricultural'). It will reveal, for instance, how much precision drops as recall for 'wildfire' is increased. Second, it helps in setting initial, plausible cost values for the subsequent CSROC model. By observing the natural trade-offs, one can make more informed judgments about the relative severity of different errors, forming the basis for the cost matrix.

Third, **Step 2: Formulation of the Cost Matrix**. Based on the diagnostic insights from the PR-curves and the known operational priorities, a cost matrix must be formally defined. This is the core of the CSROC approach. Drawing from analogous fields like medical triage and fraud detection, where misclassification costs vary significantly, the

matrix assigns a specific cost to every possible outcome ⁹¹ ⁹⁶. For this application, a plausible structure would be:

- **Correct Prediction Cost ($C_{correct}$):** Set to 0.
- **Error Costs (C_{error}):** These should be differential. A 'wildfire' misclassified as another class should have a very high cost (e.g., 100). A 'mining' misclassified as 'industrial' might have a medium-high cost (e.g., 50), while a 'wildfire' misclassified as 'agricultural' might have a medium cost (e.g., 40). This reflects the varying levels of threat and consequence.
- **Rejection Cost (C_{reject}):** This cost should be carefully calibrated to represent the estimated cost of human review. It must be high enough to discourage unnecessary rejections but low enough to allow the model to defer confidently when uncertainty is high. This value could be informed by estimates of analyst time and the strategic importance of the task. The overall goal is to define a matrix that reflects the real-world stakes of the decision.

Fourth, **Step 3: Implementation of the Reject-Option Mechanism.** Given the exclusion of conformal prediction due to complexity, a practical implementation involves a dependent rejector ¹⁰². A common and effective method is to use the model's maximum predicted probability as a confidence score. An instance is rejected if this maximum probability falls below a certain threshold. The optimal threshold is found empirically by evaluating the model on a held-out validation set. For each potential threshold, the expected cost is calculated using the formula:

$$E[Cost] = \sum_i P(x_i) \cdot \min_{c \in \{predict(c), reject\}} Cost(decision, true_label(x_i))$$

where the sum is over all validation instances. The threshold that yields the lowest overall expected cost is selected. This process directly implements the CSROC principle in a computationally efficient manner. CatBoost's ability to handle categorical variables and its support for custom objectives also suggest it could be adapted for more integrated rejection strategies, though this would be more complex to implement ²³⁶²⁴⁹.

Fifth, **Step 4: Validation, Reporting, and Dashboard Integration.** The final system must be rigorously validated on a held-out test set that was not used in training or threshold selection. The report for the demo should go beyond standard accuracy metrics and present a suite of operational trust signals. Key metrics to report include the final achieved abstention rate (to demonstrate compliance with the workload constraint), the precision and recall for each class (especially 'wildfire'), and the estimated reduction in total cost compared to a baseline classifier that never rejects. For the dashboard,

visualizations are key. A plot showing the relationship between the confidence threshold, the corresponding abstention rate, and the expected cost would be highly effective. This would allow an evaluator to see that the chosen threshold operates within the desired operational envelope. To meet the auditability requirement, the dashboard narrative must include a dedicated section explaining the cost-sensitive framework, presenting the finalized cost matrix, and justifying the cost assignments based on the diagnostic analysis and the high-stakes nature of the mission.

Synthesis and Strategic Recommendation for a National Security Context

In conclusion, the selection of a thresholding strategy for a high-stakes satellite anomaly detection system is a critical decision that extends far beyond standard machine learning performance metrics. The analysis of Precision-Recall-based thresholding and Cost-Sensitive Reject-Option Classification reveals that the latter is the strategically superior approach for this specific application, given its explicit integration of operational constraints, risk management, and workload considerations. While the former offers valuable diagnostic insights, it is fundamentally a descriptive tool for post-hoc tuning, whereas the latter provides a prescriptive, decision-theoretic framework for proactive, optimized policy-making. For a national security agency, where the integrity of automated systems and the clarity of their decision-making processes are paramount, the rigor and audibility of CSROC make it the more defensible choice.

The primary recommendation is to adopt a hybrid workflow that uses Precision-Recall analysis as a foundational diagnostic tool and builds upon it with a fully implemented Cost-Sensitive Reject-Option Classification framework. This path acknowledges the practical constraints of a tight development timeline while delivering a solution that is technically robust, operationally sound, and tailored to the unique demands of a defense-adjacent decision-support dashboard. The process begins with calibrating the CatBoost model's probabilities to ensure that the confidence scores are reliable indicators of truth¹⁷⁸. Following calibration, a multi-class Precision-Recall analysis should be conducted to understand the model's inherent performance trade-offs and to inform the construction of a meaningful cost matrix¹³⁴³²⁹. This matrix is the heart of the CSROC system, translating abstract operational goals—such as minimizing the risk of missing a wildfire or controlling reviewer workload—into concrete, quantitative terms^{91 102}. The final implementation, using a dependent rejector based on the model's maximum predicted

probability, provides a practical and efficient method for determining which predictions to delegate to human analysts ¹⁰².

This recommended approach directly addresses all facets of the user's request. It prioritizes the management of false-negative risk for the most critical class ('wildfire') by allowing for the explicit weighting of such errors in the cost function. It respects the implicit constraint on reviewer workload by treating the cost of human intervention as a tunable parameter in the optimization process, enabling the establishment of an "abstention budget" ¹⁸⁷. Most importantly, it satisfies the stringent requirement for auditability. The decision to abstain is not a black-box phenomenon but is instead grounded in a transparent cost-benefit calculation, the logic of which can be documented and defended. This transparency is essential for building trust with a non-ML expert evaluator and aligns with the principles of responsible AI deployment advocated by bodies like NIST, which emphasize documentation, transparency, and accountability ¹²⁹ ¹³³. By framing the solution within this decision-theoretic context, the project moves beyond simply building a classifier and instead constructs a comprehensive decision-support system, a distinction that is vital for success in a high-stakes public-sector environment.

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